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Examining the mediating role of employee creativity: a study of how knowledge management influences and employee performance works in higher education institutions

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Abstract

Higher education institutions (HEIs) endeavor to increase employee performance (EP) by fostering employee creativity (EC) and knowledge management (KM) in order to gain and maintain a competitive edge. While taking into account the user's viewpoint on the impact of knowledge management (KM) processes—more especially, knowledge creation, retention, dissemination, and utilization—in HEIs, this study critically reviews the body of current literature. The study primarily examines how creative motivation may improve productivity among employees and how employee creativity can act as a mediator in the interaction between knowledge management and employee performance. A structured survey was administered to 432 academic staff members from India's research-intensive HEIs, and SPSS and AMOS were used for statistical analysis. The results show that employee performance is significantly impacted by knowledge management both directly and indirectly through employee inventiveness. In order to maximize knowledge-driven innovation and productivity, academic administrators and policymakers can benefit greatly from the insights, this study provides on the dynamic interactions among knowledge management, employee creativity, and institutional performance in HEIs.

Keywords Employee creativity, Employee performance, Innovation, Knowledge management, Higher education

Introduction

Higher education institutions (HEIs) depend on collective effort, partnership, and diligence, which can considerably influence Knowledge Management (KM) and better affinity and analysis [87](#), resulting in enhanced employee performance (EP). HEIs are encouraged to produce, obtain, preserve, disseminate, and leverage knowledge Ahmaad et al. [2](#), Iqbal et al. [57](#) to attain this objective. HEIs have transformed and reframed in research and economic

development, surpassing outdated and stagnant teaching practices. Currently, HEIs are seen as institutions for enthusiastic and innovative knowledge creation and dissemination [19](#). Higher education institutions (HEIs) have a crucial role in the processes of generation, acquisition, sharing, storing, and application of knowledge, which in turn contribute to the progress and development of society and the economy. Efficient knowledge management can facilitate the progress and enhancement of HEI facilities and advancements, such as teaching, learning, research, curriculum development, management, and strategic development [57](#). Researchers have examined and analyzed the knowledge management capabilities of organizations, specifically in the context of knowledge

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management and knowledge management enablers [40]. Knowledge management encompasses various activities that involve the creation, acquisition, storage, sharing, and utilization of knowledge, which ultimately contribute to the competitive advantage of higher education institutions (HEIs), as stated by Baarley et al. 20. The enablers of knowledge management include various factors that are instrumental in promoting significant advancements in knowledge management [102]. Prior studies have established that knowledge management enablers have a solitary or instantaneous favorable effect on the effectiveness or productivity of enterprises 92. By scrutinizing the current literature on knowledge management and educational research, scholars have been able to pinpoint noteworthy deficiencies that necessitate further investigation. Successful deployment of a knowledge management system (KMS) can confer a competitive edge to a company vis-à-vis its rivals [103], Mcafadyen and Canella 75. Employees possess a distinctive set of knowledge, competencies, and experiences that can provide organizations with a competitive advantage, thereby rendering their knowledge a strategic asset [39]. According to Janeet et al. [58], knowledge management takes place organically within organizations, with employees being invariably implicated in this process (knowledge generation, knowledge dissemination, knowledge utilization) to varying degrees, and the extent of their knowledge contribution hinges on the opportunities afforded to them within the organization [49]. The explores how variables within the knowledge management (KM) process help manufacturing firms minimize innovation risk, using resource-based views and organizational learning theory. Through PLS-SEM and fsQCA analysis, the findings confirm a strong link between green implementation, KM practices, and innovation risk reduction, offering valuable insights for scholars and industry experts 6.

Employee performance can be measured using objective, subjective, or mixed/quasi-measures, and there are various definitions available in the literature. Enabling knowledge creation can improve employee creativity and lead to the development of new products, ultimately increasing sales and profitability for the organization, as per Richard et al. [88] and Lee and Choi [66], Zhu et al. 110. Employee performance now relies on the knowledge they possess, and the more they know, the more proactive they can be for the organization (Kianto et al. [63], Ma et al. [71]). Employees' intentions and behavior are interconnected and critical for organizational success. The employees' intentions are closely linked to their knowledge-related tasks, which not only affect their performance but also contribute to the overall achievements of the knowledge management system and the organization's performance 33. Despite the growing importance

of knowledge management (KM) in HEIs due to the abundance of knowledge-based resources, there are still significant challenges in implementing effective knowledge management policies [26, 108]. The current policies are either inadequate or unpredictable, indicating the need to further investigate the factors contributing to successful knowledge management 37. Despite the vital role of knowledge and knowledge management in universities, these institutions are often not recognized as knowledge-intensive organizations and lack proper knowledge management strategies [94], Iqbal et al. 57. Furthermore, research on knowledge management in HEIs in developing economies is limited [86, 93]. The study of Indian HEIs is particularly important because the sector is still in its early stages 34, Turner and Aacker 104, and the rapid growth of higher education has shifted the balance of power in the region 91, Yuiying et al. 109. Furthermore, the increasing competition between HEIs, fueled by a surge in student enrollment, makes it imperative to examine knowledge management in Indian higher education [93]. Knowledge management (KM), employee innovation, and performance in higher education institutions have all been the subject of recent research. Mazhar and Akhtar's 74 study looked at the connection between knowledge management and creativity among Lahore university instructors. The results showed a strong positive relationship between KM practices and teacher creativity, indicating that good KM may create an atmosphere that supports creative teaching methods. Alzghoul et al. [11] investigated at how knowledge-oriented leadership and strategic thinking affect staff creativity in private colleges in Jordan. According to the study's findings, knowledge-oriented leadership further moderates the association between strategic thinking and employee creativity. This suggests that university employees' creativity may be greatly increased by leadership philosophies that prioritize information exchange and strategic insight. The significance of manufacturing companies may reduce supply risk by implementing knowledge management (KM) procedures and green technologies (GTA). Data from 203 Zhengzhou businesses were analyzed using PLS-SEM and fsQCA. The results show a high correlation between supply risk reduction, KM procedures, and GTA, offering guidance for sustainable supply chain management 7.

Compared to other sectors, knowledge management in Indian higher education is still in a nascent stage 92. Therefore, studying the impact of trust and organizational climate as facilitators of knowledge management and their influence on employee performance through organizational learning in India is of significant importance. The research aims to address the knowledge gaps in the field of knowledge management in HEIs by

examining the relationship between trust, organizational climate, and various aspects of knowledge management to encourage employee performance, Darroch 30, Shahzaad et al. 95. The study also aims to investigate whether any intervening factors affect the association between knowledge management and employee performance 61. This will help in identifying the mechanisms through which knowledge management impacts the performance of HEIs. Given the scarcity of research on knowledge management's role in Indian HEIs, this study is poised to make a significant contribution toward comprehending its function in these establishments. By underscoring knowledge as a vital organizational asset that enhances employee performance, the study aims to provide theoretical insights into the existing literature on KM.

Knowledge management, or KM, is becoming more widely acknowledged as a crucial component of improving organizational performance. However, its use at higher education institutions (HEIs), particularly in developing nations like India, is still insufficient and poorly understood. While HEIs produce and share information, research shows that they frequently lack organized knowledge management plans, which makes it difficult to effectively use intellectual resources for staff and institution development [94], Iqbal et al. 57. A more methodical approach to knowledge generation, distribution, and usage is also required due to the increased rivalry among institutions brought about by India's higher education sector's explosive growth [93]. However, there are still a lot of unanswered questions about how organizational climate and trust support knowledge management and how these elements eventually affect worker performance through organizational learning. Although earlier research highlights the importance of knowledge as a strategic asset [39], further research is required to understand the intervening factors that influence the relationship between knowledge management and employee performance, especially given the distinct sociocultural and economic environment of Indian HEIs. Institutions may find it difficult to maximize their knowledge management (KM) capacities if these knowledge gaps are not filled, which might affect their capacity to promote innovation, staff productivity, and overall institutional competitiveness. By examining the link between trust, organizational environment, and knowledge management at HEIs and how it affects employee performance, this study hopes to add to the body of knowledge in academia and provide useful information for policymakers.

Based on the current research gaps, the project aims to answer the following research topics:

RQ1 Does knowledge management directly affect employee performance in India HEIs?

RQ2 Can employee creativity serve as a mediator of the correlation between KM and EP in HEI located in India?

Significant research gaps still exist despite the growing emphasis on knowledge management (KM) in higher education institutions (HEIs), especially when considering emerging nations like India. However, there is still much to learn about their relevance to Indian HEIs. Corporate organizations or HEIs in developed economies are the subject of the majority of KM research [37, 93]. Considering KM adoption at Indian HEIs is still in its infancy, there is a dearth of empirical data about the effects of KM practices on worker performance in this setting. This study closes the gap by focusing on KM implementation in Indian HEIs, taking into account their particular difficulties, quick growth, and escalating competitiveness. Many Indian HEIs lack clear knowledge management (KM) strategies [108], and the policies that are in place are frequently inconsistent or poorly executed 26. In order to help policymakers and institutional leaders create more successful knowledge management frameworks, this research identifies important success factors and obstacles to KM adoption. Trust and organizational environment have not been thoroughly examined as important facilitators of KM success at HEIs, despite the fact that previous research recognizes the relevance of KM enablers [102]. This study looks at how these elements affect knowledge management (KM) processes (creation, sharing, and usage) and how that affects worker performance.

The effect of knowledge management (KM) techniques on employee performance at higher education institutions (HEIs), especially in India, has been further investigated in recent research. Research by Sivasubramanian and Jeyaraj 98 looked at how faculty members' work performance in a few HEIs in Coimbatore, Tamil Nadu, related to different knowledge management approaches. The results imply that improving teaching staff commitment, attachment, and work participation requires good knowledge management strategies. Sadiq et al. [90] looked at the variables influencing academics' knowledge-sharing practices in Indian HEIs. Key factors impacting knowledge-sharing behavior were found by the study, offering insights into how educational institutions might encourage faculty cooperation and information sharing. Research conducted in [90] examined the variables influencing academics' knowledge-sharing practices at Indian HEIs. The study found important factors affecting faculty members' knowledge-sharing behavior,

offering guidance on how educational institutions might encourage a collaborative and information-sharing culture.

Formulation of hypotheses and establishment of the research framework

The theoretical foundation of this investigation is based on two works, namely Gold et al.'s [40] model of knowledge management capability and Grant's [43] knowledge-based view (KBV). Gold et al.'s model of knowledge management capability comprises two elements: the capability of knowledge infrastructure and the capability of knowledge management. By this model, the effectiveness of an organization relies heavily on the capability of its knowledge infrastructure, which establishes a conducive environment for knowledge management and results in the continuous enhancement of employee performance. Grant's approach to knowledge management emphasizes the advantages that employees gain from upgrading their knowledge and prioritizes knowledge assets that are rare, unique, and valuable. Drawing on these hypothetical standpoints, this study propositions an incorporated paradigm to investigate the correlation between knowledge management, employee creativity, and employee performance. The proposed framework will be used to test the research hypotheses and provide insights into the interplay between these variables in Indian HEIs.

Knowledge management and employee creativity

According to multiple studies, it is thought that creativity is not just an individual cognitive activity, but also a social and communication process supported by knowledge management. The works of [23, 106] utilize cognitive elements, such as learning theory and knowledge management, to showcase how reciprocity, trust, and communication among members of communities of practice and informal networks can encourage knowledge sharing, learning, and creativity. Burt's [25] mechanical void assumption is a noteworthy study that employs a knowledge management perspective to explore creativity and its relationship with social networks, offering evidence of their impact on individuals.

In conformity with [77, 97], the possession or availability of domain-relevant knowledge enhances the possibility of achieving creative performance. The stance is also backed by Obstfeld [81], who asserts that open networks are more effective in generating novel concepts. Moreover, research indicates that creativity is predominantly a product of collaborative endeavors within teams and that group situations tend to boost employee creativity [89]. Consequently, the emphasis has transitioned from the creative abilities of an individual to

the network of connections they possess. Significantly, attention is centered on how these connections aid in the transfer and management of knowledge, which ultimately results in innovative outcomes (Andreeva and Kianto 14). This approach to fostering creativity through knowledge management is consistent with the literature on knowledge management and social media, as discussed earlier. Grant [42, 107] proposed a conceptual interactionist model that investigates the impact of group dynamics on individual creative behavior. However, more recent research has focused mainly on assessing the impact of group dynamics on collaborative creativity, as well as identifying organizational and procedural factors that could affect the outcomes of collaborative creativity [85]. The impact of corporate social responsibility (CSR) on employee creativity in Pakistan's hospitality sector, emphasizing the mediating roles of happiness and creative self-efficacy and the moderating role of polytonicity. Using SMART-PLS analysis on data from 409 hotel employees, the findings reveal that CSR fosters creativity, supporting the UN-SDGs by enhancing emotional and cognitive processes that drive sustainable and innovative practices.

The findings of these studies have been inconsistent due to the utilization of diverse techniques [51] to evaluate various creativity outcomes, including patents, managerial performance, managerial creativity engagement, and knowledge dissemination. As organizations face increasing global competition, the need for continuous creativity and renewal has become a top priority [22, 32, 48, 101]. Lucía Muñoz-Pascual and Jesús Galende [78] proposed a different perspective for studying the correlation between knowledge management and creativity, Karim et al. [62] proposed examining the behavioral aspect of knowledge management readiness, whereas Mariano and Awaazu [73]. Research on the mediating function of creativity and knowledge management as a business result should be conducted empirically, according to Adreva and Kiante [14] and Heisig et al. [52]. Based on the ideas of componential creativity and social exchange, green transformational leadership cultivates employee green creativity through employee green passion and green atmosphere. The results, which were obtained using PLS-SEM on 464 workers in the US manufacturing sector, demonstrate that green transformational leadership fosters green creativity and that this link is strengthened by green perceived organizational support. These findings offer important insights for businesses striving for sustainable innovation [1].

H1 Knowledge management is positively related to significance.

Knowledge management and employee performance

Knowledge management and employee performance are closely intertwined as knowledge is considered one of an organization's most valuable resources [4]. It can be found in the minds of employees and documents, routines, clients, and suppliers. Data processing leads to the transformation of data into information, which in turn becomes knowledge when it is understood, applied, and practiced by employees [72]. Through repeated application, this knowledge becomes ingrained in the individual's mind and can be observed through their experiences, skills, and problem-solving abilities [36]. Organizations recognize knowledge management as a crucial element to improve their performance [13]. This study investigates at how knowledge management techniques affect worker performance in Malaysia's hotel sector, emphasizing the mediating effects of capacity building and organizational commitment. The results of PLS-SEM analysis and survey data from 291 workers demonstrate that good knowledge management improves worker performance, highlighting the significance of encouraging dedication and a learning culture for long-term success [83].

They invest in educating, training, collecting, storing, and spreading information throughout all levels of the organization to cultivate new knowledge and leverage existing knowledge to achieve performance excellence [96]. Using a desktop research design, examine how knowledge management (KM) affects worker happiness and performance. Effective knowledge management (KM) improves productivity, lowers job stress, and creates a positive work environment. The findings highlight the necessity for firms to match KM with leadership, organizational culture, and long-term strategic goals in order to achieve the best possible results for their employees.

Consistent with Jaradat et al. [59], knowledge management impacts employees by influencing their learning, which refers to the continuous change in behavior. Learning is a process that enables individuals to acquire information and skills through interaction with social and cultural systems in their environment [67].

H2 Knowledge management is positively related to employee performance.

Employee creativity and employee performance

In the findings of Arling and Chun's [15] research, holding interdepartmental meetings is considered a productive approach to fostering the creation and dissemination of new knowledge by enabling employees from diverse domains to exchange their viewpoints. Nold [80] found

that organizing interdepartmental sessions for employees from various fields to exchange their viewpoints can enhance the implementation of a knowledge management system in an organization. Similarly, Lloria and Ortiz [70] examined Nonaka's knowledge creation model and identified several enablers that facilitate knowledge creation, such as the intention of the company or employee to participate in the process. According to Muñoz-Pascual and Jesús Galende [78], organizations that prioritize creativity tend to have a heightened awareness of market conditions and are better equipped to adjust to changes, resulting in the dissemination of innovative ideas and organizational learning that ultimately enhances long-term performance. Shahzad et al. [95], effective management of knowledge management can enhance employee creativity, which in turn can lead to improved employee performance. Despite the importance of the employee's mind as the locus of knowledge creation, it has been neglected in previous years in favor of the organization's routines, hindering the creation of new knowledge. According to Kalpic and Bernus [61], the effective utilization of knowledge resources by employees is crucial for generating value. Therefore, this study focuses on the role of employees in knowledge creation, sharing, and utilization, rather than solely on the organization's systems. Creativity, As per Craft's [27] assertion, creativity is a productive exercise that generates innovative and worthwhile results. Numerous studies, including those by (Haenain et al., [44]; Liu & Genge, [68]), have demonstrated that employee creativity has a significant positive impact on performance. Employees who possess a creative inclination are internally resilient and unique, ultimately leading to improved performance.

With an emphasis on variables including job experience, education, and perceived system validity, this study investigates the influence of creativity, knowledge, and motivation in the creative recommendations made by frontline hospitality personnel. The results, which are based on data from 153 hotel workers in Tenerife, show that innovation is greatly influenced by creativity, experience, and faith in recommendation systems, with leisure hotels encouraging more suggestion-driven innovation than business hotels [100]. The magnitude and velocity of creativity are factors that contribute to higher performance returns [41] in addition to this, the results discuss various types of creativity, including process, technology, and administration, which lead to improved employee performance. Mehralian et al. [76] conducted a recent study that indicated the importance of meeting customer demands with creative products to achieve company performance and profitability. The rapid advancement of technology and the increasing competition have forced organizations to re-evaluate their means of gaining a

competitive edge and prioritize creativity and knowledge, as stated by Lerro et al. [67].

H3 Employee creativity is positively related to employee performance.

Knowledge management and employee performance and employee creativity

This research proposes an integrated model of knowledge management from the user's perspective to enhance employee performance. It expands on the theoretical model presented by Christina and Chang [28] and highlights the role of employee creativity as a mediator between knowledge management intention and employee performance. The objective of this research is to investigate knowledge management in the Indian higher education institution setting from the viewpoint of the users, taking into account various antecedents and moderating factors 11. The relationship is present between knowledge sharing (KS) and innovative performance (IP) in Pakistan's manufacturing sector, with employee creativity (EC) as a mediator. Using survey data from 158 managers and regression analysis, the findings confirm that KS positively influences IP, with EC playing a significant mediating role, highlighting the importance of fostering creativity to enhance innovation in emerging markets 105

This study suggests that higher education professionals should note employees' willingness to engage with knowledge management. According to research, having favorable intentions and conduct toward creating, sharing, and utilizing knowledge can result in the development of innovative products and enhanced employee performance. Bagheri et al. [18] propose that knowledge management can establish a conducive learning atmosphere in organizations, thereby promoting creativity, while Al-Hakim and Hassan 5 suggest that this can ultimately lead to improved performance. The study examines the moderating effect of an information-intensive culture on the link among knowledge processes, employee innovation, and business performance. The results of a survey conducted among 197 workers in Pakistan's service industry show that knowledge application, sharing, and acquisition improve company performance. Employee creativity mediates this effect, and a knowledge-intensive culture amplifies the influence of knowledge processes on creativity 56.

The impact of knowledge management processes on employee performance, emphasizing the mediating role of employee creativity in a healthcare setting. Using SEM analysis on data from 218 healthcare professionals in the Hail Health Cluster, the findings confirm that knowledge

sharing, utilization, and discovery enhance creativity, which in turn improves employee performance, highlighting the need for effective knowledge management strategies in healthcare organizations. Knowledge management can influence employee creativity and performance. To fully comprehend the relationship between knowledge management and employee performance, the outcome of creative learning should be evaluated. Thus, in future studies investigating the association between knowledge management and employee performance, employee creativity should be carefully considered [69, 79].

H4 EP mediates the relationship between EM and EC.

Underpinning theory

The componential theory of creativity (Amabile, [12]) offers a paradigm for comprehending how employee creativity (EC) in higher education institutions (HEIs) mediates the relationship between knowledge management (KM) and improved employee performance (EP). Intrinsic motivation, creativity-relevant processes, and domain-relevant skills combine to produce creativity. By developing expertise, promoting teamwork, and establishing an inventive workplace culture, knowledge management (KM) stimulates creativity. Employees provide innovative answers to difficult problems through knowledge management (KM) exercises including information sharing and collaborative learning, which enhances work performance, productivity, and institutional efficacy. This demonstrates how important EC is in converting KM initiatives into improved EP within HEI.

3. Research methodology

The "10-times rule" approach [45] is a popular minimum sample size estimation technique in PLS-SEM that is grounded on the idea that the sample-to-variable ratio of 15:1 or 20:1 is ideal, but it shouldn't be less than 5:1 when taking the sample-to-variable ratio into account [46]. The methods for applying Cochran's [29] formulae to determine sample size for continuous and categorical variables are described in this publication.

A total of 432 data points were collected, with a margin of error. Random stratified sampling was the sampling technique used.

Sample

There were 432 responders in all, according to the demographic information gathered from Uttarakhand's higher education institutions (HEIs) as shown in (Table 1). Of these, 41.7% are women (180) and 58.3% are males (252). The age distribution shows that 25.5%

Table 1 Description of participants' demographics (Source Self complied)

Demography	Particular	Frequency
Gender	Men	252
	Women	180
Age	26 to 40 yrs	207
	41 to 55 yrs	115
	56 to 70 yrs	110
Qualification	Post graduate	235
	PhD	121
	Post PhD	129
Sum		432

of people are between the ages of 56 and 70 (110), 47.9% are between the ages of 26 and 40 (207), and 26.6% are between the ages of 41 and 55 (115). In terms of credentials, 29.9% have a post-PhD qualification (129), 28.0% have a PhD (121), and 54.4% have a postgraduate degree (235). This data highlights a highly qualified academic workforce in Uttarakhand's HEIs.

Constructs used

Knowledge management (KM)

KM was measured using an adapted version of the scale developed by Gold et al. 40. The construct included items designed to assess the processes of knowledge creation, sharing, and utilization within the organization. A total of 15 items were used, each rated on a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree).

Examples of items include:

- "Employees in my institution frequently share knowledge with one another."
- "The institution provides systems to facilitate knowledge retention and retrieval."

The scale demonstrated strong internal reliability, with a Cronbach's alpha value of 0.91, confirming its robustness.

Employee creativity (EC)

Employee creativity was assessed using a scale adapted from Zhou and George [111]. This scale focuses on employees' ability to generate innovative ideas, solve problems creatively, and propose novel approaches to work tasks. The construct was measured using 4 items, rated on the same 5-point Likert scale.

Examples of items include:

- "I am encouraged to suggest new and innovative solutions to work challenges."
- "My institution fosters a supportive environment for experimentation and creativity."

The reliability of this scale was also strong, with a Cronbach's alpha value of 0.87.

Employee performance (EP)

Employee performance was measured using a scale adapted from Kirkman and Rosen [65]. This scale evaluates both individual and team contributions to organizational goals. A total of three items were included, rated on a 5-point Likert scale.

Examples of items include:

- "I consistently meet the performance targets set by my institution."
- "My work contributes significantly to achieving my institution's objectives."

The scale had a Cronbach's alpha value of 0.89, confirming its reliability for measuring performance outcomes.

4. Analysis and results

Description of participants' demographics

The participants for this current research were academic and other employees aged 25 to 60 who worked in HEIs all over India. Male respondents made up 58% of the total respondents, while female respondents made up 42%. The table provides a thorough profile of the respondents.

Results

Descriptive statistics

Table 2 presents the descriptive statistics, which provide evidence supporting the following conditions: normality, absence of multicollinearity (with all VIF values below 10), strong reliability and interitem consistency for all constructs, and no presence of autocorrelation.

This table shows the tests of normalcy (Kolmogorov–Smirnov and Shapiro–Wilk) for three variables—creative engagement (CE), knowledge management (KM), and employee performance (EP)—based on a sample size of 432. Both the Shapiro–Wilk and Kolmogorov–Smirnov tests determine if the data has a normal distribution.

All variables have *p*-values (Sig.) of 0.000, which is less than the conventional significance level of 0.05. This indicates that the data for EP, KM, and CE do not follow a normal distribution, hence rejecting the null hypothesis of normality in Table 3.

Since both tests provide significant findings, the dataset looks to be non-normally distributed,

Table 2 Descriptive statistics (Source SPSS AMOS)

Descriptive statistics												
	N	Range	Minimum	Maximum	Mean		Std. deviation	Variance	Skewness		Kurtosis	
	Statistic	Statistic	Statistic	Statistic	Statistic	Std. Error			Statistic	Std. Error	Statistic	Std. Error
KM	432	4.00	1.00	5.00	3.0167	.05712	1.18719	1.409	.061	.117	-.631	.234
EC	432	4.00	1.00	5.00	2.9514	.05864	1.21891	1.486	.093	.117	-.797	.234
EP	432	4.00	1.00	5.00	2.9769	.06205	1.28969	1.663	.024	.117	-1.048	.234

Sign ** $p < 0.01$, * $p < 0.05$.**Table 3** Normality test (Source SPSS AMOS)

Tests of normality						
	Kolmogorov–Smirnov ^a			Shapiro–Wilk		
	Statistic	Df	Sig.	Statistic	df	Sig.
EP	.123	432	.000	.928	432	.000
KM	.103	432	.000	.933	432	.000
EC	.109	432	.000	.935	432	.000

Table 4 Model fit (Source SPSS AMOS)

Measure	Estimate
CMIN	246.454
DF	206
CMIN/DF	1.196
CFI	0.995
NFI	0.969
RFI	0.965
IFI	0.995
TLI	0.994
CFI	0.995
PClose	1

*** $p < 0.001$

even though the Shapiro–Wilk test is often more trustworthy for smaller samples.

4.2.2 Analytical approach

The paper used confirmatory factor analysis for evaluating the model in this study using AMOS 23.0. According to Hu and Bentler's [55] and Hair et al.'s [45]

criteria, the model fit is deemed exceptional, as indicated by the statistics presented in Table 4.

The construct validity of the study was assessed using the AMOS 23.0 program, which employed the following criteria: average variance extraction values (AVEs) greater than 0.5, correlation ratios (CR) greater than AVE, and the square root of the AVE being greater than the correlation among other structures, as outlined by Hair et al. [45]. The results of Table 5 indicate that there were no concerns regarding the construct validity of the study. It was important to examine the biases because only one approach was used to obtain the data. The Gaskin-proposed common method bias test was conducted using the common latent factor approach [38]. In Table 6, this is accomplished by including the CLF latent factor in our measurement model, which matches all the observed variables. Two sets of standard regression weights were used: one with and another without CLF. The very small variation in weights (< 0.200) demonstrated that there was no bias. Table 7 lists the factor loading of every variable associated with constructions. The convergence of the variables with constructs is confirmed by all loadings being greater than 0.7.

Table 5 Model validity (Source SPSS AMOS)

	CR	AVE	MSV	MaxR(H)	KM	EC	EP
KM	.969	.676	.249	.969	.822		
EC	.886	.66	.232	.886	.482	.812	
EP	.883	.716	.249	.884	.499	.447	.846

Table 6 Total variance explained (Source SPSS AMOS)

Total variance explained									
Component	Initial Eigenvalues			Extraction sums of squared loadings			Rotation sums of squared loadings		
	Total	% of variance	Cumulative %	Total	% of variance	Cumulative %	Total	% of variance	Cumulative %
1	11.908	54.126	54.126	11.908	54.126	54.126	9.997	45.441	45.441
2	2.410	10.952	65.079	2.410	10.952	65.079	3.256	14.798	60.239
3	1.578	7.174	72.253	1.578	7.174	72.253	2.643	12.014	72.253

Table 7 Factor loading (Source SPSS AMOS)

			Estimate
KM15	←	KM	
KM14	←	KM	20.177***
KM13	←	KM	20.553***
KM12	←	KM	21.579***
KM11	←	KM	21.082***
KM10	←	KM	19.746***
KM9	←	KM	20.711***
KM8	←	KM	20.358***
KM7	←	KM	21.077***
KM6	←	KM	20.546***
KM5	←	KM	20.628***
KM4	←	KM	20.666***
KM3	←	KM	20.209***
KM2	←	KM	20.658***
KM1	←	KM	20.692***
EC4	←	EC	
EC3	←	EC	17.87***
EC2	←	EC	18.247***
EC1	←	EC	17.911***
EP3	←	EP	
EP2	←	EP	19.672***
EP1	←	EP	20.289***

Hypotheses research

After the assessment of the basic assumptions (normality, absence of multicollinearity, model fitness, reliability, and validity), the hypotheses were assessed using SEM (Fig. 1). Further, the bootstrapping method was used to test the mediation effect in the suggested model. Findings of the path analysis used to verify the suggested hypotheses are shown in Table 8.

According to hypothesis 1, EP in HEIs is highly correlated with KM. The KM's effect on EP is quantified by the beta coefficient of 0.386 ($t = 7.670$, $p\text{-value} < 0.001$). This proves the first hypothesis. Similarly, the second and third hypotheses were substantiated, as shown in Table 8. The findings of the experiment to test Hypothesis 4, which proposed EC mediates the

link among both KM and EP, are shown in Table 8. The mediation effect is confirmed if the indirect effect is significant [53]. It is necessary to first take into account the requirements for mediation as proposed by Hayes and Preacher [50]. According to Hayes and Preacher (2013), the presence of a significant direct effect signifies the mediation effects. Further, the presence of a significant direct effect followed by a significant indirect effect confirms partial mediation. It demonstrates how, in contrast to (Baron and Kenny's [21]) method, the substantial mediating variable is the sole criterion for mediators.

Table 9 shows the presence of a mediator ($ab = 0.242^{***}$). Thus, the mediation effect is confirmed, validating hypothesis four. Further, there is a significant direct effect. This supports the notion that EC plays a partially mediating role in the correlation between KM and EP in Figure 2.

- Common method bias (CMB): As data were collected via self-reported questionnaires, the risk of CMB was mitigated using Harman's single-factor test and the common latent factor (CLF) approach. Minimal variations in factor loadings (<0.2) indicated negligible CMB.
- Non-response bias: Steps were taken to reduce non-response bias by offering flexibility in response formats (online/offline) and by conducting follow-ups. The final response rate was approximately 70%, indicating a robust participation level.
- Selection bias: Stratified random sampling ensured representation across organizational roles, reducing the likelihood of overrepresentation or underrepresentation of specific groups.
- Cultural bias: Items were adapted and pre-tested through a pilot study with 30 participants to ensure cultural appropriateness and relevance within the Indian HEI context.
- Social desirability Bias: Anonymity was guaranteed to respondents to encourage honest and unbiased responses. Additionally, reverse-coded items were included to detect and minimize socially desirable answers.

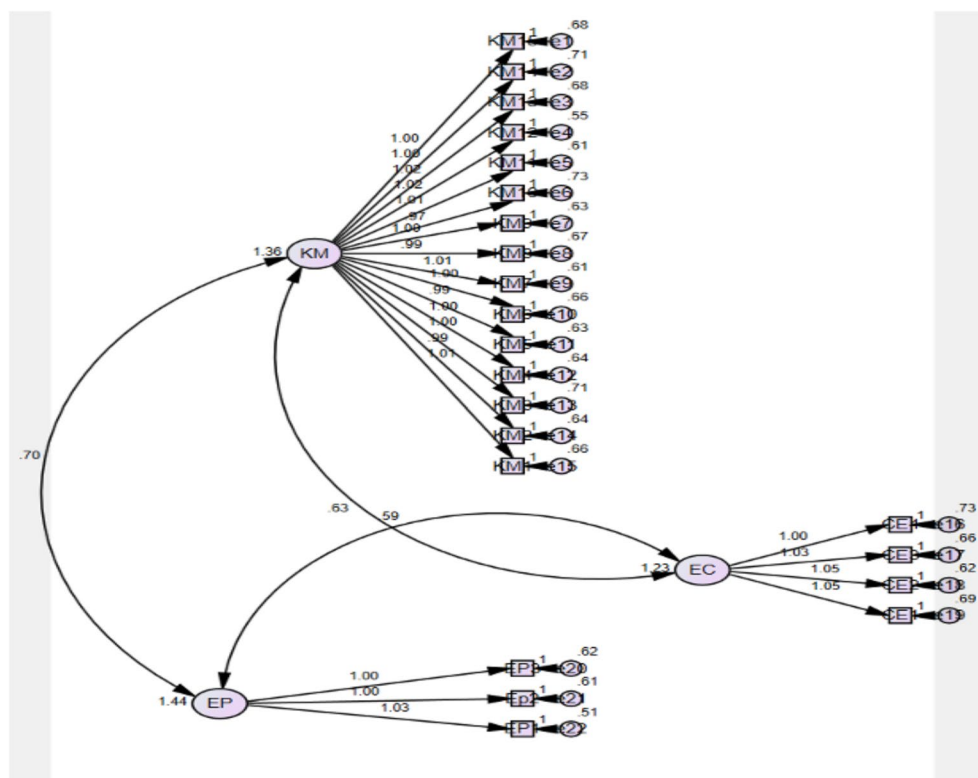


Figure 1 Measurement model (Source SPSS AMOS)

Table 8 Result of path analysis (Source SPSS AMOS)

Hypotheses	Analysis	S.E.	C.R.	Result
EP ← KM	.386	.050	7.670***	H1 Accepted
EC ← KM	.461	.044	10.421***	H2 Accepted
EP ← EC	.252	.049	5.144***	H3 Accepted

*** $p < .001$

Table 9 Mediation result (Source SPSS AMOS)

Hypotheses	Direct impact	Indirect impact	Total impact	Finding
KM → EC → EP	0.010¥	0.242	0.252***	H4 accepted

*** $p < .01$; * $p < .05$; ¥ $p > .05$.

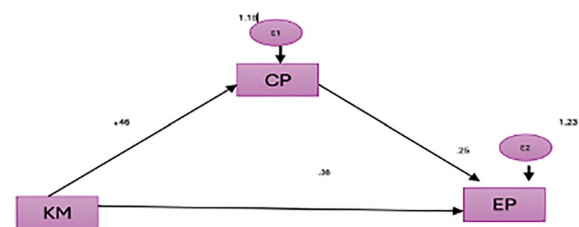


Figure 2 Structure model (Source SPSS AMOS)

Discussion

The objective of this research paper is to evaluate the extent of the association between knowledge management (KM) and employee performance (EP) and to explore how employee creativity (EC) practices can shed light on the underlying mechanisms of this

correlation. The initial hypothesis posits a positive correlation between KM and EP. In Table 10, the beta coefficient of 0.386 ($t = 7.670$, p -value 0.001) in Table 5 provides evidence supporting this hypothesis, indicating a significant impact of KM on EP. This finding is

Table 10 Validation of hypothesis (Source SPSS AMOS)

Hypothesis	Regression weight	Result	Conclusion
H1	0.461***	Accepted	KM significantly and positively impacts EP.
H2	0.252***	Accepted	KM significantly and positively impact EC
H3	0.386***	Accepted	EC significantly and positively impacts EP.
H4	.242***	Accepted	EC significantly, and KM and EP

*** $p < 0.01$

consistent with the conclusions drawn by Dirani [31], Bryson [24], and Jiménez-Jiménez and Sanz-Valle [60]. Sulistyanto et al. [99] conducted a thorough literature analysis and found that knowledge management (KM) has a beneficial impact on employee performance. They also highlighted KM as a crucial component of organizational success and a useful tool for learning companies. Building on this framework, Alshammary and Ali [10] looked at how knowledge management (KM) procedures, particularly knowledge exchange, usage, and discovery, affected worker performance in a medical environment. Their results showed that the link between KM procedures and performance is mediated by employee creativity, indicating that performance outcomes may be improved by encouraging creativity through efficient KM tactics. Alzghoul et al. [11] looked at how knowledge-oriented leadership and strategic thinking affected the creativity of staff members at private institutions in Jordan. According to the results, knowledge-oriented leadership and the development of a strategic thinking culture may greatly increase staff creativity, which will boost HEI performance.

Knowledge management fosters an environment where employees possess the ability to identify, analyze, and organize knowledge effectively. Such a culture in organizations also facilitates greater independence among employees. These factors' impact, individually and collectively, significantly influences employee performance. The second hypothesis proposed a relationship between KM and EC. The beta coefficient of 0.461 ($t = 10.421$, p -value 0.001) in Table 5 indicates the significance of KM on EC. Thus, Hypothesis 2 is supported. The study confirms O'Dell and Hubert's [82] findings by identifying KM as a favorable predictor of the EC process. Employees high in KM feel aligned with EC initiatives. This fosters innovation and the emergence of new ideas. Hypothesis 3 proposed that EC and EP have a positive relationship. The significant strength of the EC on EP is quantified by the beta coefficient of 0.252 ($t = 5.144$, p -value 0.001) in Table 5. Thus, the study confirms the findings of Alavi [8], Muhammed et al. [77], and Hand Arrington [47] by identifying EC as a favorable predictor of the EP process (2003).

The results are consistent with those of Alavi & Leidner [9], who highlighted that knowledge management (KM) creates an atmosphere in which workers are better able to recognize, evaluate, and arrange knowledge, which enhances job performance, problem-solving skills, and decision making. Additionally, O'Dell and Hubert [82] emphasized that companies with robust knowledge management cultures encourage increased employee independence and flexibility, which strengthens the observed beneficial impact on EP. Our findings go counter to those of King et al. [64], who proposed that the impact of KM on EP differs greatly depending on the industry and organizational structure. Our results show a clear and substantial association, indicating that even in knowledge-intensive institutions like HEIs, KM can be a powerful predictor of EP. In contrast, their study concluded that the success of KM depends on leadership support and technical infrastructure.

Furthermore, [111] discovered that innovative workers are more proactive, involved, and able to come up with original ideas, all of which improve job performance. Our results similarly support those of Richards et al. [88], who showed that companies that support EC gain a competitive edge, increase productivity, and foster innovation.

Our results, however, contradict those of Baer and Oldham [16], who proposed that the nature of work and organizational support systems determine how creativity affects performance. They discovered that unless workers are given enough resources and acknowledgment, innovation could not necessarily translate into improved performance. This disparity emphasizes the necessity of more investigation into the contextual elements affecting the EC-EP interaction.

Hypothesis 4 proposed that EC be the mediator in the connection between KM and EP. The relationship between KM and EP through EC (path $ab = 0.242^{***}$) in Table 9 indicates the presence of a mediator. The result found a substantial indirect effect; the direct relationship between KM and EP (path $ca = 0.01$) is not noteworthy. This supports Hypothesis 4 and establishes the prerequisite for partial mediation. This is in agreement with the conclusions reached by Bacer and Sinkla [17], Pesbmaa et al. [84], Honhming et al. [54].

Theoretical contribution: By offering empirical proof of KM's direct impact on EP and EC in the setting of higher education institutions (HEIs), this work builds on earlier studies. This study emphasizes the value of knowledge management (KM) at academic institutions and how it improves faculty and staff performance, whereas earlier research mainly concentrated on business settings.

Recommendation: Policymakers should combine creativity-driven initiatives with knowledge management (KM) tactics to improve staff performance at higher education institutions (HEIs). Teaching and research productivity may be improved by establishing organized knowledge-sharing platforms, promoting multidisciplinary cooperation, and cultivating an innovative culture. Performance evaluation methods, professional development initiatives, and incentives should all be designed to acknowledge creativity as a major performance driver. Additionally, KM policies may be improved for greater institutional efficiency through data-driven decision making using analytics and AI tools. By putting these tactics into practice, HEIs may foster an atmosphere that is innovative and knowledge-rich, which will eventually enhance academic achievement and staff performance.

Implications

The main goal of the current study is to give suggestions, offer recommendations, and provide assistance to educationists, practitioners, and policymakers to transform and revitalize HEIs in India. The study offers ideas for future theoretical and empirical research as well as a theoretical framework. The work has theoretically improved the body of knowledge on KM, EP, and EC. It has advanced fresh insights and knowledge of these relationships by advancing current studies into the Indian scenario. The uniqueness of the study is that it has attempted to provide a framework for institutional effectiveness from two perspectives: individual-level KM and organizational-level EC. The study emphasized the nurturing of KM of individuals and strengthening of EC in the institutions to boost EP. These findings also offer suggestions for enhancing KM and facilitating EC, which are important predictors of EP. The study also aims to caution educationists, practitioners, and policymakers associated with HEIs of the potential challenges. It suggests stressing KM and EC for the enhancement of EP that will ultimately lead to organizational survival, growth, and prosperity. This will boost the performance of their organizations. The study also provides practical implications for HR professionals, managers, administrators, and leaders of HEIs for creating KM and EC for EP.

Practical implications: The results highlight how important it is for HEIs to have organized knowledge management systems to encourage innovation and boost

worker productivity. Institutions ought to concentrate on: improving knowledge-sharing procedures via training courses, online databases, and cooperative platforms. Fostering an innovative culture by giving staff members independence, creative flexibility, and credit for their original ideas, as well as incorporating knowledge management (KM) into professional development programs, performance evaluations, and strategic decision-making to maximize staff performance.

Conclusion

The objective of this study is to verify the proposed model by examining how employees' intentions toward knowledge management affect their performance in a higher education institution (HEI) while also exploring the mediating role of creativity. By conducting this research, the study aims to provide validation for the proposed model and contribute to the existing knowledge on the subject. The research seeks to explore how these factors interrelate and influence the overall performance of employees in the given context. By analyzing the data, the study aims to provide insights into the effectiveness of knowledge management strategies in enhancing employee performance in HEIs. The practice of knowledge management encompasses various activities such as the creation, storage, sharing, and utilization of knowledge, all of which are aimed at improving organizational performance. Furthermore, knowledge management can also contribute to the improvement of employee performance by creating a conducive learning environment that promotes creativity and innovation. By leveraging knowledge management strategies, organizations can gain a competitive advantage and achieve sustainable growth. Employees who require frequent updates on knowledge and possess creativity are considered to be knowledge-intensive employees. These individuals play a crucial role in the competitiveness and survival of an organization. Their ability to constantly update their knowledge base and think creatively allows them to adapt to changing circumstances and develop innovative solutions to problems. As such, organizations that value knowledge-intensive employees are more likely to remain competitive and thrive in their respective industries. Additionally, these employees often serve as key contributors to the development of new products or services, which can further enhance the organization's competitive position. This study possesses two significant strengths. Firstly, it comprehensively explains the relationship between knowledge management, employee creativity, and employee performance. The study delves deeply into the subject matter, leaving no stone unturned in its quest to provide a thorough understanding of the topic. Secondly, the data collected in this study are specific to job categories within

higher education institutions. This specificity means that the findings are particularly relevant to faculty members of universities who require creativity to enhance their performance and foster creativity.

The study's focus on higher education institutions is also noteworthy, as it provides insights into this sector's unique challenges and opportunities. Overall, this study is a valuable contribution to the field of knowledge management, offering insights that can be applied in a variety of organizational contexts. Additionally, the findings of this study hold practical significance in terms of utilizing knowledge management effectively to foster employee creativity and improve overall performance. The study's insights can be applied in various organizational contexts, helping businesses to develop strategies that foster innovation and creativity among their employees. Additionally, the study provides valuable ideas for future research in this field, highlighting areas that require further exploration to deepen our understanding of the relationship between knowledge management, employee creativity, and performance. By shedding light on these critical issues, this study contributes to the advancement of knowledge management theory and practice, enabling organizations to achieve their goals more effectively.

Limitations of the study

With the fact that the study "Employee Creativity as a Mediating Factor in the Relationship between Knowledge Management and Employee Performance: An Analysis of Higher Education Institutions" clarifies the complex relationships that exist between knowledge management, employee creativity, and performance in higher education settings; it is crucial to recognize its limitations. First, the study's exclusive emphasis on higher education institutions may limit how broadly the findings may be applied to other sectors or companies. Furthermore, response bias might be introduced by the study's dependence on self-reported data from employees since participants can give socially desirable rather than accurate responses. Because the study design was cross-sectional, proving a relationship between factors may be more difficult, necessitating longer term research to draw more reliable findings. Finally, even though employee creativity is identified in the study as a mediator, other possible moderators or mediators that could also impact the link between employee performance and knowledge management may be missed. Accordingly, to give a more thorough grasp of the topic, future research initiatives could overcome these constraints.

Scope for future work

Although KM in Indian HEIs has been examined in this study, little is known about how cultural factors—such as power distance and collectivism versus individualism—affect KM procedures and how they interact with EC and EP. Examine how KM practices and systems are modified in HEIs that operate in culturally varied regions, such as Scandinavia or East Asia (e.g., China and Japan). Comprehending cultural differences might improve global knowledge management tactics and guide region-specific HEI procedures. There is a lack of research on how modern information management technologies, such as AI-powered knowledge repositories, collaborative platforms, and predictive analytics, may promote EC and enhance EP in HEIs. Examine how HEIs in various locations might improve information sharing and innovation by integrating cutting-edge technology, including cloud-based knowledge management systems, big data analytics, and machine learning. Insights into how the dynamics of KM-EC-EP at technologically advanced HEIs are affected by digital transformation. There has not been much focus on how financial mechanisms, governmental regulations, and socioeconomic circumstances influence knowledge management techniques at HEIs across various areas. Examine the effects of policy frameworks and resource availability on KM-EC-EP relationships in HEIs in developed and developing nations. Assisting in the formulation of policy suggestions to improve KM efficacy in HEIs with limited resources.

Abbreviations

KM	Knowledge management
EC	Employee creativity
EP	Employee performance
PLS	Partial least squares
HEI	Higher education institution
SEM	Structural equation modeling
VIF	Variance inflation factor
CFA	Confirmatory factor analysis
AVE	Average variance extracted

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Megha Ojha was involved in introduction and literature review and data collection and conclusion. Dr Vinay Kandpal was performed concept of the topic and all over correction of the work. Archana Singh was done data analysis.

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Consent for publication

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